

Smart Weather Alert System - Version 1.0

Specification

Tagline: Stay Safe Before Weather Changes

Project Goal

Build a single-page weather alert application that detects user location, fetches live weather, predicts dangerous conditions, and sends browser/mobile notifications.

Navbar

Notification Bell shows a red indicator only for high-priority alerts. User icon opens profile with Name, Email, City, Coordinates, Notification toggle, Change Location, Logout.

Weather Dashboard

Dynamic location, temperature, humidity, wind, pressure, visibility, weather condition and weather icons from Weather API.

24 Hour Forecast

Dynamic hourly forecast with icons and temperatures.

Active Alert

Green state: No Active Alerts. Red state: Severe weather with warning message.

Upcoming Alerts

Future alerts such as Heavy Rain, Strong Wind, Thunderstorm, Heatwave and Flood.

Notification Flow

Weather API -> Spring Boot Scheduler -> Rule Engine -> Firebase Cloud Messaging -> Phone/Browser Notification.

Priority Rules

Green = Safe, Yellow = Moderate, Red = Severe.

User Setting

Notify before: 15 Minutes, 30 Minutes, 1 Hour, 2 Hours.

Database

Users, Alerts, Notification Logs with FCM Token storage.

Tech Stack

HTML, CSS, JavaScript, Java Spring Boot, Firebase Auth, Firestore, Firebase Cloud Messaging, Geolocation API, Weather API.

Roadmap

1. Folder Structure
2. UI
3. Weather API
4. Location API
5. Firebase Auth
6. Firestore
7. Java Backend
8. Scheduler
9. Notifications
10. Deployment

Version 1 Scope

Single page responsive UI, live weather, dynamic icons, profile management, upcoming alerts and push notifications.